
WHITE PAPER · WORK, ADOPTION & JUDGMENT

Felt Productivity Is Not Measured Productivity

Why AI adoption should be judged on delivered output, not on how fast it feels

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EXECUTIVE SUMMARY

Before scaling an AI tool, compare its delivered output with a baseline (from before the tool, a held-back group, staged waves, or work records), and keep self-reported speed as a separate record.

People can name a number for how much faster AI made them. In a careful trial that checked such a number against a clock, it ran well ahead of the finished work, and nothing shows which way it errs in your workflow. A program scaled on that number is scaled on sentiment. A three-part standard replaces it: a baseline, an output measure that heavier tool use cannot inflate, and a review that keeps felt speed and delivered output in two columns.

- 01** Treat a self-reported speedup as an estimate of how using the tool felt: in one randomized trial, developers believed AI sped them up 20 percent while the clock showed 19 percent slower, and a later study was too unreliable to settle the question.
- 02** Leave adoption charts, prompt counts, and survey speedup out of the evidence; they show that people use and like the tool, not that the output improved.
- 03** If the tool is already live, hold a comparable group back, roll out in waves, or rebuild the before-picture from records and call it an estimate; with none of these, the output reads “not measured” and expansion waits.
- 04** When the team feels faster but delivered output has not moved, that gap is the finding the review exists to catch, not an anomaly to explain away.

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They felt faster. The clock said otherwise.

Ask people how much faster AI made them and they can give you a number. That number is an estimate of how the work felt, not a measurement of what it delivered, and the two can be far apart. A leader should treat it as an estimate, judge the tool on the output it delivered against a baseline taken before the tool arrived, and keep the feeling in a separate record. Here is the finding behind that advice. In a 2025 randomized controlled trial, the research nonprofit METR had sixteen experienced open-source developers complete 246 real tasks on their own software projects. Each task was randomly assigned to be done with or without AI help. Going in, the developers expected the tools to speed them up by 24 percent. Coming out, having lived through the work, they still believed AI had sped them up by 20 percent. The clock said the opposite. They took 19 percent longer with the tools than without.[1]

EXHIBIT 1

Capable people misread their own speed

19%

In METR's randomized trial, experienced developers took 19 percent longer with AI, yet afterward still believed it had sped them up by 20 percent — the gap this standard governs.

METR randomized controlled trial: 16 experienced open-source developers, 246 real tasks, July 2025.

What the trial shows is narrow, and the limits matter. It does not show that AI makes people slower. It measured sixteen expert developers, on mature projects they knew well, using early-2025 tools, at one moment in time. Change any of those circumstances and the number can move. Read it as a verdict on the tools and you claim more than one small trial can support. What it does show is one documented case. Capable people believed one thing about their own speed, the clock recorded another, and the belief survived direct contact with the evidence.

METR has since measured again with newer tools. A second study, begun in August 2025, estimated that AI-allowed tasks took 18 percent less time for the ten

developers who returned and 4 percent less for 47 new ones, and both ranges include no change. METR says this data is an unreliable signal, mainly because developers who did not want to work without AI left the study, which likely pulls the estimate toward less speedup. It believes developers are likely more sped up now than in early 2025, and calls its own data only very weak evidence for how much.[2] So do not quote the early-2025 figure as a rate. It remains the only trial this paper cites that checked a feeling against a clock, and there the feeling was wrong.

A second source looks like a match and is not. The DevOps Research and Assessment (DORA) program at Google Cloud surveyed nearly 3,000 working professionals for its 2024 report. Respondents who used AI more reported higher individual productivity, flow (sustained focus), and job satisfaction, and the report's delivery findings pointed the other way: AI adoption "negatively impacts software delivery stability and throughput." [3] Both sides of that split come from what respondents answered. DORA's productivity measure is a score for how effective and efficient people feel, and its delivery measures are respondents' answers about the application they work on. DORA therefore does not set a feeling against a measurement. It shows two sets of answers, one about the individual and one about the team's delivery, that did not move together. The delivery answers also changed within a year: the 2025 report found AI adoption now linked to higher throughput, with delivery instability still rising.[4]

Taken together, the sources support a narrower claim than "people overrate AI." In one randomized test, the feeling ran ahead of the clock. In METR's later attempt, the answer was unclear and probably different. In the 2024 DORA survey, answers about the individual and about delivery diverged, and the delivery answers then changed. A leader can conclude that the feeling is not a reliable stand-in for the output, and that no one can tell from outside which way it errs in a given workflow. That is the problem a governance standard has to handle, and it is why this paper asks for a measurement.

Why the feeling is a weak basis for the decision

METR's February 2026 follow-up says that developers' self-reported speedups "can be quite unreliable." [2] That is not a claim that people lie. One plausible reason, which the trial did not test, is that the tool answers instantly. The draft appears. The blank page fills. That sensation can feel like progress even when checking, correction, and re-prompting follow and erase the saving.

Set the forecast beside the result. A 24 percent expected gain is optimism about a new capability. The 20 percent believed gain is the more telling figure. It was reported after the work was done, when experience might have corrected the estimate, and it did not. This paper's reading is that a survey asking employees how much AI helped may capture enthusiasm, tool preference, and the natural wish to have spent one's effort well. The trial did not test that explanation. It is a reason not to let a survey stand in for a record, and the same caution applies to the people a workplace survey reaches, who may be the ones who chose or champion the tool.

The second reason is getting stronger. METR's follow-up reports that the "wider adoption of AI has made it more difficult to measure task-level productivity".[2] When surveyed, 30 to 50 percent of developers said they chose not to submit some tasks because they did not want to do them without AI.[2] The more normal AI use becomes, the harder it is to find unaided work to compare against. In this paper's reading, that makes the number that is always available, what people say about themselves, more tempting to use. An organization scaling AI on that number alone is scaling on sentiment, with no way to tell from the survey whether it is too high or too low. No portfolio sets out to fund sentiment. It can end up doing so when no one built the alternative.

What to measure instead: three things

The tool is new. The way to measure a change in how work gets done is not. The U.S. National Institute of Standards and Technology (NIST) makes measurement one of the four core functions of its AI Risk Management Framework. Its MEASURE function uses "quantitative, qualitative, or mixed-method" tools to analyze and monitor AI risk and related impacts.[5] Where trade-offs among trustworthy characteristics arise, "measurement provides a traceable basis" to inform management decisions.[5] Measurement works here as a control, not as an after-the-fact report. It lets a leader decide whether to use the tool more, hold, or stop.

So the standard this paper proposes is narrow and testable. Three things have to exist before an AI-assisted workflow is being governed on evidence. A baseline, a before-picture captured before the tool changed the work. A work-output signal that reflects delivered value rather than activity. A review that reads the felt experience and the delivered outcome as two separate records. They work as a sequence. Without a baseline there is nothing to compare against. Without an output signal the comparison counts the wrong thing. Without the review the comparison changes no decision. A program can show a rising adoption chart and a glowing internal survey

and still not produce those three. It is then being run on impression, however disciplined the rollout looked.

Take the before-picture first

A CIO magazine story on the return on AI investment quotes a former global CIO at a retailer describing her own company's AI rollouts: "We didn't start with a baseline." [6] That is one practitioner's account, not a measure of how often it happens, but it names the problem. Without a before-picture, no after-picture can be checked. Any improvement can be credited to the tool. Any decline can be explained away. There is no honest record of how the work performed when a person did it unaided. The baseline does not need to be elaborate. It needs to be captured before the workflow changes, on the same measure the review will use afterward. A baseline rebuilt from memory after adoption inherits exactly the optimism this problem is made of.

If the tool is already live and no baseline exists

The tool may already be in use by the time anyone asks for a baseline. There are three ways to get a comparison, listed in this paper's judgment from strongest to weakest. No source tests them against each other.

- Hold a group back. Keep a comparable team, site, or queue on the old way for a set period and compare the same output signal. The comparison is fair only if the held-back group resembles the others, so do not let people choose which group they join.
- Roll out in waves. If the tool has not reached everyone, turn it on for one group at a time and compare groups before the next wave. One company introduced an AI assistant to 5,179 customer-support agents in stages and measured issues resolved per hour, an output count. [7] That is one workflow at one company. It shows the method, not what to expect.
- Rebuild the before-picture from records. Pull the same output measure for the months before the tool from the work-tracking records: cases closed, cycle time, rework. Call it an estimate and say why. Staffing, volume, season, and definitions may have changed over the same months. A baseline rebuilt from anyone's memory does not count.

If the tool has already reached everyone, only the third route is left. If none of the three is possible, the output column of the review reads "not measured." That is different from a weak result. A weak result is evidence that the tool is not paying off yet. A missing one means the organization does not know. Expansion then waits, unless the organization would still choose it if the gain were zero.

Count finished work, not activity

A baseline is only as good as what it measures, and this is where an AI dashboard can go wrong. It counts activity: seats provisioned, prompts sent, tokens consumed, features accepted. Activity is easy to count, and it rises as more people use the tool. The SPACE framework argues against that reflex. Researchers at GitHub, Microsoft, and the University of Victoria published it in ACM Queue. Developer productivity, they argue, is about “more than an individual’s activity levels” and “cannot be measured by a single metric or dimension.”[8] SPACE names five dimensions: satisfaction and well-being, performance, activity, communication and collaboration, and efficiency and flow. For a workflow outside software, the useful move is to ask which of the five an AI metric actually touches. Adoption charts and prompt counts are activity. Survey speedup is satisfaction. Only performance and, sometimes, efficiency describe whether the work came out better, and those two are the hardest to capture, so a dashboard can easily lack them.

A work-output signal answers a question about delivered value: cycle time on a real case, rework rate, error rate, coverage, or throughput of finished units that met standard. It is chosen so that using the tool more cannot move it by itself. Usage is an input. Output is what the portfolio funded. Adoption rate, active-user counts, prompts per employee, and self-rated speedup belong on an adoption dashboard, where the question is whether a rollout reached people. They stay out of the evidence a portfolio uses to decide whether to deepen use, because none of them can tell a workflow that got better from one that merely got busier.

A contributed opinion column in CIO magazine, written by a technology-company field CTO, puts the same point at the level of the task: the governance machinery of “steering committees, approval gates and KPI dashboards” governs the organization chart, while “the value, or the leak, happens inside the workflow, at the level of the actual task the AI is doing.”[9] It is an opinion column, not research, and it is cited for the distinction, not as proof. The same CIO magazine story, reporting on one law firm’s AI tools, says there is “no straight line between the investment in that tool and increased revenues”.[6] That is why a gain on one task may not reach the bottom line.

Keep the feeling and the result in separate columns

The trial points most directly to this third element. The review keeps two columns and never collapses them. One holds what people report about the experience of the work. The other holds what the work actually did against the baseline. Both are legitimate. Satisfaction and flow are real, they matter for keeping people and for adoption, and DORA’s respondents reported more of both as they used AI more.[3] They are recorded as experience, though, next to the output signal, not merged into

it. The felt column informs adoption and retention. The decision to expand rests on the output column. When the two columns agree, use can grow with confidence. When they diverge, the team feels faster while the delivered work has not moved, or has slipped, and that divergence is the most valuable thing the review produces. It is precisely the case a satisfaction survey alone would have scored as a win.

This paper asks one question: does what people report about their speed match what the work delivered? “Measure AI by the workflow it changes” covers which measures a whole AI workflow should carry beyond that, and “Benefits realization is where transformations quietly fail” covers following a benefit after launch.

What it costs to scale on the wrong number

No source here prices these costs, so they are this paper's reasoning about what follows when a scaling decision rests on a feeling. The first is misdirected capacity. The tools and workflows that produce the strongest feelings attract the next round of investment, whether or not they produced the strongest results. Scarce integration, change, and checking capacity follows the enthusiasm rather than the evidence, and the organization compounds its bets on the wrong thing because it felt best.

The second cost is credibility, and it works slowly. A program promoted on a promised speedup eventually meets the results it was supposed to move. When the delivered numbers do not show the gain the survey implied, the gap reads as a broken promise, not a measurement error, and it spends the sponsor's credibility along with the wider appetite for AI investment.

The third cost arrives later. Savings claimed on felt speed are sometimes cashed in advance, in a headcount plan, a benefits case, or a budget cut justified by an efficiency no one measured. A saving that existed only as a feeling still has to be paid back, in rework, escalation, and quiet rehiring, once the work fails to deliver it. Measuring the work is likely cheaper than any of these, and cheapest before the commitment, not after the shortfall.

How the review works when the columns disagree, and when they agree

Consider a team that adopts an AI assistant across a recurring, high-volume workflow, the kind where a small saving per case would add up. This is an illustration of how the standard would run, not an account of any organization, and its results are

invented to show the two outcomes. Suppose adoption is fast and the sentiment is excellent. The internal survey reports a large felt speedup, and the case for expanding the license writes itself. Under an impression-led process, that is where the decision gets made.

Under this standard, the survey is one column, and it is not the column that decides. The review pulls the output signal the team agreed on before rollout: cycle time on completed cases that met the quality bar. It compares that against the baseline captured from the pre-tool period. Either result is possible, which is the reason to keep both columns. In reviews I have seen, the two columns can disagree, and a team cannot know which case it has without both.

First suppose they disagree. The felt speedup is genuine, and the measured cycle time on completed, standard-meeting cases has barely moved. The measurement does not say by itself where the saving went, only that it did not reach the delivered outcome. That is the finding. A real, local time saving stopped somewhere short of the result the portfolio funded, and only the output signal made the shortfall visible. Where it stopped is the workflow owner's question to run down. The disagreement is no reason to abandon the tool. The review is doing its job. It changes the next decision from "expand it, because people love it" to "find out where the saving stopped, then measure again." The felt speed told the team something true about the experience of the work. The output signal told them something true about the delivery of it.

Now suppose the columns agree. The team reports a genuine speedup, and the output signal on completed, standard-meeting work has moved with it against the baseline. Then the review produces the opposite verdict, and produces it with evidence a skeptical finance partner can inspect. That is the case for deepening use, expanding the license, and moving the workflow from trial to standing operation. Separating the columns does not mean distrusting every good feeling. It makes the yes as well evidenced as the no.

Objections to the standard, and what to do about each

"People will work to the output number." Once funding hangs on a measure, people can improve the measure without improving the work: close cases faster at lower quality, or split one job into several. This risk cannot be removed. It can be reduced by choosing a measure of finished units that met the quality bar, so speed cannot be bought with rework, by reading a sample of the work itself and not only the count,

and by tying the measure to the workflow and its owner, not to any individual's pay or ranking.

"Measuring costs too much." It costs something, and no source here prices it. This paper's expectation is that a baseline rebuilt from records costs an analyst's time to define the measure and pull the data, plus the owner's time at each review. A held-back group costs the benefit that group does not get during the test, and effort to keep it comparable. A staged rollout costs little when the rollout would be staged anyway. The cost is proportionate when the standard is applied where spending is large or a decision to scale is near. For a small tool, record the survey result as experience only and claim no gain. If no one funds the measuring, the only number that arrives free is the self-report, so an unfunded standard reverts to it. A KPMG AI leader quoted in the same CIO story says the measurement of how AI use correlates with work output is "fundamentally lacking".[6] That is one practitioner's view of the gap, not a measure of it.

"We already have the data, because the tool logs everything." The logs record activity: who used the tool, how often, and for how long. They do not record whether finished work improved, and logs that follow individuals' activity carry a surveillance cost and can erode trust. The standard needs a different number, not more watching: a small set of output signals at the level of the workflow, tied to a baseline and owned by someone accountable for that workflow.

"The result will be out of date before we use it." It may be. The tools change, and so can the answer: METR now says developers are likely more sped up than a year ago, on very weak evidence,[2] and DORA's throughput finding changed direction between its 2024 and 2025 reports.[3][4] Treat a result as dated to the tool version and the workflow it measured, and measure again at the next decision to expand.

The one-page record

The three-part standard is the whole standard. The page below records it for one workflow, so that it survives a change of sponsor and a scaling decision finds the evidence already defined. It adds no fourth test. It adds the facts the three parts need in order to be used: who owns the result, how the baseline was obtained, what decision follows, and who pays for the measuring. The workflow owner writes it before the decision to expand, or before rollout when the tool is new, and revises it at each later decision. It is a living page, not a one-time approval record. A leader's request is simple: do not approve an expansion until the owner has produced this page.

EXHIBIT 2**One page records the standard for a workflow, its owner, and its cost**

- 1 Workflow and owner**
The recurring workflow and the one person accountable for its delivered outcome.
- 2 Baseline and its source**
Output measure from before the tool, or from a held-back group, waves, or records.
- 3 Output and felt records**
A measure heavier use cannot move, beside what people report, in separate columns.
- 4 Decision**
Deepen, hold, redesign the workflow, or stop; a missing measure holds the decision.
- 5 Cost**
Who funds the measuring, and for how long.

A standard like this does not slow adoption. It protects it. It lets a leader say yes to deeper use on evidence, and lets them redirect rather than abandon a tool whose value is leaking inside the workflow. It is aimed at one failure: rolling a capability out across an organization on the strength of how fast it felt.

What these studies do and do not show

The METR trial is randomized but small, and it licenses no general claim that AI reduces productivity; this paper makes none. METR's later study estimated that AI-allowed tasks took less time, with wide ranges, and METR itself calls that estimate unreliable because of who left the study.[2] DORA's findings carry a different limit. They are associations across survey respondents, not a controlled experiment, and both the individual and the delivery results are what respondents reported. The 2024 and 2025 reports disagree on throughput.[3][4] The staged rollout this paper cites is one workflow at one company.[7] SPACE and NIST supply the vocabulary of measurement, not evidence about AI,[5][8] and the two CIO magazine pieces are practitioner voices, a news feature and an opinion column, cited as such.[6][9]

EXHIBIT 3

What the evidence establishes, and what it does not

- **METR trial**
A small randomized trial: self-reported speedup ran well ahead of the clock.
- **METR follow-up**
Newer tools, wide ranges; METR calls its own estimate unreliable.
- **DORA 2024 and 2025**
Survey answers on both sides; the throughput finding changed between reports.
- **SPACE and NIST**
The vocabulary of measurement, not evidence about AI.
- **Not established**
A universal metric, review schedule, cost, or best route when no baseline exists.

The deeper caution comes from METR's own follow-up. Wider adoption is making task-level measurement harder.[2] The clean baseline this standard depends on is getting more expensive to capture, not less. An organization that waits may find the unaided comparison has vanished from its own operations. That is an argument for building the measurement now, while a before-picture, a held-back group, or a rollout in waves is still possible, rather than a reason to defer it until the tools stop changing. No source here establishes a universal work-output metric, a required review schedule, a benefit multiplier for AI, what measuring costs, or which route without a baseline works best. Those are properties of a specific workflow, and the standard is deliberately a method for finding them rather than a set of numbers to adopt.

The test that matters

The test of AI governance is not the enthusiasm of a rollout or the confidence of a survey. A capable tool can produce both, and early. The real test comes when the felt speed and the delivered outcome disagree: can the organization see the

disagreement, and act on the outcome? An enterprise that measures the work, with a baseline, a real output signal, and a review that keeps the feeling and the result in separate columns, is governing its AI. An enterprise that scales on how fast the work feels is governing its mood, and funding it as if it were evidence.

Felt productivity is real, and it is worth having. It is simply not the same thing as measured productivity, and only one of the two can tell a portfolio where its money went.

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